MUHAMMAD BASITH K

Software Engineer | Python Full Stack Developer

+91-9048884058 — basithkdrbk@gmail.com

GitHub — LinkedIn — HackerRank — Portfolio

Professional Summary

Software Engineer with hands-on experience in Python Full Stack Development using Django, Django REST Framework, React.js, JavaScript, and PostgreSQL. Skilled in building responsive web applications, developing RESTful APIs, implementing secure authentication, and designing scalable backend solutions. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, SQL, Git, and modern software engineering practices, with hands-on experience developing real-world full-stack applications.

Technical Skills

Programming Languages

Python, JavaScript (ES6+), C++, SQL

Frontend

React.js, HTML5, CSS3, Bootstrap 5, Tailwind CSS

Backend

Django, Django REST Framework (DRF), REST APIs, JWT Authentication

Databases

PostgreSQL, MySQL

Developer Tools

Git, GitHub, Postman, VS Code, Render, Jupyter Notebook

Core Concepts

Object-Oriented Programming, Data Structures & Algorithms, MVC Architecture, CRUD Operations, Authentication & Authorization, API Integration, Responsive Web Design, Version Control

Experience

Python Full Stack Developer Intern

Gen Corpus Data Hub — Kozhikode, Kerala

Jan 2026 – July 2026

- Developed and maintained full-stack web applications using **Python, Django, Django REST Framework, React.js, JavaScript, and PostgreSQL**.
- Designed and implemented RESTful APIs, integrated frontend and backend modules, and optimized database queries to improve application performance.
- Built responsive user interfaces with React.js and collaborated with backend development using Django REST Framework following clean architecture principles.
- Implemented authentication, CRUD operations, API integration, debugging, testing, Git-based version control, and participated in feature development throughout the software development lifecycle.
- Worked in an Agile development environment while collaborating with senior developers to deliver scalable and production-ready web applications.

Professional Projects

JobSphere – AI-Powered Career & Recruitment Platform

Technologies: React.js, Django REST Framework, PostgreSQL, Simple JWT, Bootstrap 5, Cloudinary, Vercel, Render, Neon PostgreSQL

Links: Repository — Live Demo

- Built a production-ready full-stack recruitment platform supporting **2 user roles** (Candidate and Recruiter) with secure JWT authentication and role-based authorization.
- Developed and integrated **28 RESTful API endpoints** with a responsive React frontend supporting job posting, job search, application tracking, profile management, resume uploads, and complete CRUD operations while leveraging **Cloudinary** for secure media storage.
- Built an **AI Career Hub** featuring **4 AI-powered career tools**, including ATS resume analysis, AI resume feedback, cover letter generation, and mock interview assistance.
- Deployed the application on **Vercel** and **Render** with **Neon PostgreSQL** and **Cloudinary**, delivering a scalable cloud-hosted full-stack application.

PrepPilot – Placement Preparation Platform

Technologies: React.js, Django, Django REST Framework, PostgreSQL, Simple JWT

Links: Repository — Live Demo

- Built a full-stack placement preparation platform using React, Django, Django REST Framework, PostgreSQL, and Simple JWT for secure user authentication.
- Designed and implemented **10 custom database models** supporting study planning, placement tracking, mock interviews, calendar scheduling, and user profile management.
- Developed RESTful APIs, role-based dashboards, and company-specific mock interview generation with predefined technical and HR question sets.
- Deployed the application on **Render**, enabling public access for demonstration and testing.

City Hospital Management System

Technologies: Python, Django, PostgreSQL, Bootstrap, HTML5, CSS3, Render, Gunicorn, WhiteNoise

Links: Repository — Live Demo

- Built a role-based Hospital Management System supporting **3 user roles** (Administrator, Doctor, and Patient) with secure authentication and authorization.
- Designed and implemented **5 custom database models** for department management, doctor profiles, appointment scheduling, user management, and medical report handling.
- Developed responsive dashboards, appointment booking workflows, medical report upload/view functionality, and automated email notifications for appointment confirmations.
- Deployed the application on **Render** using **Gunicorn** and **WhiteNoise**, configuring production-ready static and media file handling.

Additional Projects

Uncertainty-Aware SegFormer for Satellite Imagery (Final Year Project)

Technologies: Python, PyTorch, SegFormer (MiT-B3), Vision Transformer, Deep Learning, Computer Vision, OpenCV, Google Colab, CUDA, YAML

Links: Repository

- Developed an uncertainty-aware semantic segmentation framework by integrating Evidential Deep Learning (EDL) with the SegFormer (MiT-B3) Vision Transformer for high-resolution satellite imagery.
- Replaced conventional Softmax classification with a custom evidential head to estimate Dirichlet distributions and pixel-wise epistemic uncertainty in a single forward pass.
- Designed an optimized training pipeline using dynamic 512×512 spatial cropping, enabling efficient training of large satellite images on NVIDIA A100 GPUs while maintaining spatial alignment.
- Implemented a custom evidential loss combining Bayes Risk and Kullback–Leibler Divergence with annealing to improve model calibration and uncertainty estimation.
- Evaluated model robustness using synthetic Out-of-Distribution (OOD) corruptions and achieved approximately 1.5× faster inference than Monte Carlo Dropout while maintaining competitive segmentation accuracy.

TerraWatch – ML-Based Disaster Prediction Platform

Technologies: Python, Machine Learning, Plotly, NumPy, GIS, Raster Data Processing

Links: Repository

- Developed a geospatial analytics platform for disaster susceptibility prediction using Machine Learning and spatial datasets.
- Processed six high-resolution raster datasets and generated disaster susceptibility maps through clustering-based analysis.
- Improved usable spatial data coverage to approximately 90% by preprocessing and aligning GIS datasets.
- Built interactive 3D visualizations using Plotly to support intuitive exploration of disaster risk regions.
- Applied data engineering techniques to manage large geospatial datasets and improve prediction reliability.

Education

SRM Institute of Science and Technology, Chennai, India

Bachelor of Technology (B.Tech.) in Computer Science and Engineering

Aug 2022 – May 2026

CGPA: **7.29 / 10.0**

NM Higher Secondary School, Thirunavaya

Higher Secondary (Computer Science Mathematics)

Score: **1041 / 1200**

MES Central School, Kuttipuram

CBSE Class X

Score: **369 / 500**

Relevant Coursework

Data Structures & Algorithms	Object-Oriented Programming
Database Management Systems	Operating Systems
Computer Networks	Software Engineering
Machine Learning	Artificial Intelligence

Certifications

- AWS Academy – Machine Learning Foundations
- Oracle – Certified Foundations Associate
- Cisco – Networking Essentials
- Fortinet – Certified Associate in Cybersecurity
- Deloitte – Technology Job Simulation (Forage)

Leadership & Achievements

- Led a three-member team in developing the Uncertainty-Aware SegFormer research project for semantic segmentation of satellite imagery.
- Earned HackerRank 5 badges in C, C++, and Java along with the Problem Solving badge.
- Developed multiple end-to-end Full Stack applications using Django, Django REST Framework, React.js, and PostgreSQL.
- Experienced in building REST APIs, authentication systems, responsive web applications, and deploying Django applications.